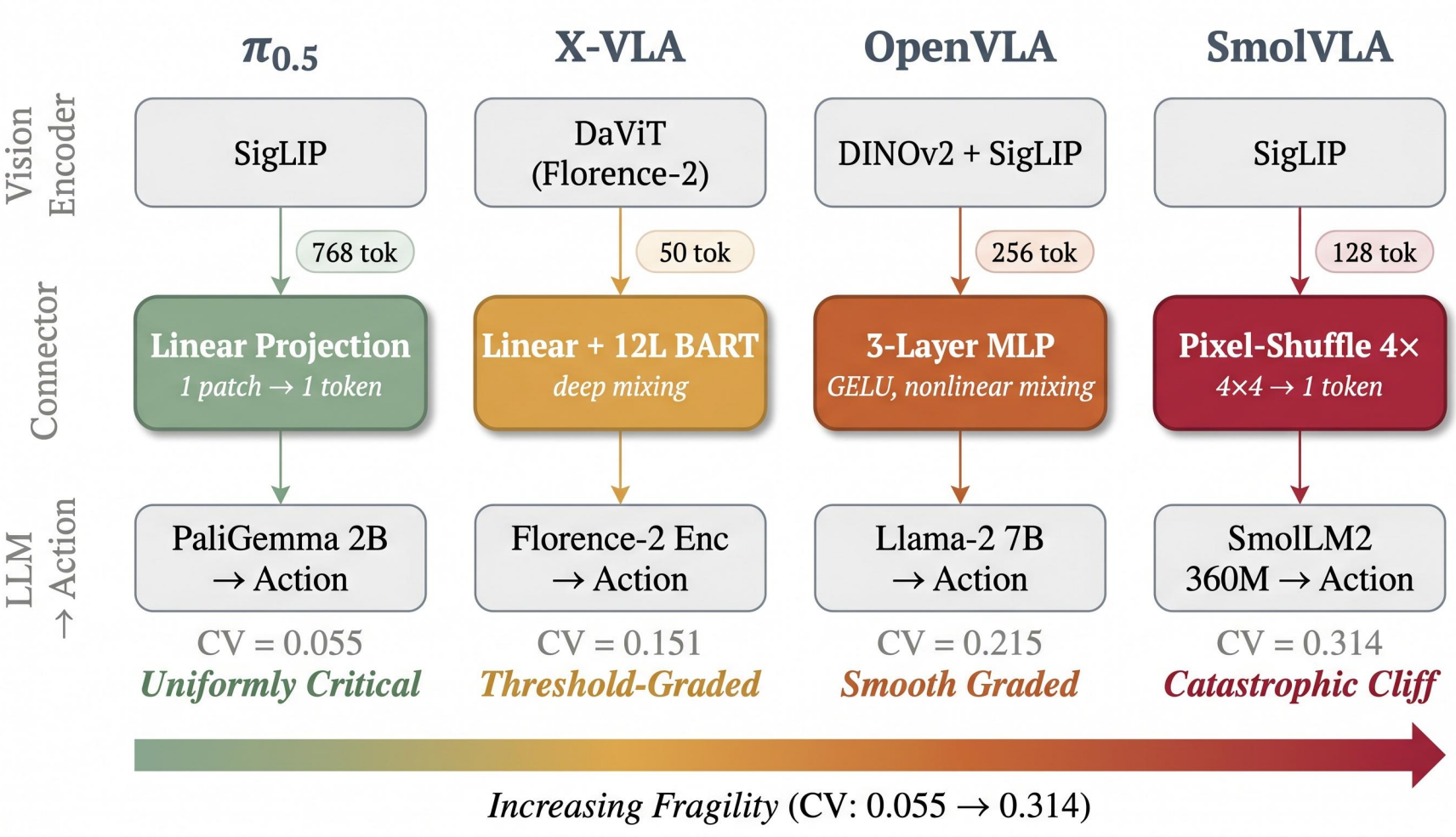

$\pi_{0.5}$
X-VLA
OpenVLA
SmolVLA
Vision Encoder
Connector
LLM → Action
SigLIP
DaViT (Florence-2)
DINOv2 + SigLIP
SigLIP
768 tok
50 tok
256 tok
128 tok
Linear Projection
1 patch → 1 token
Linear + 12L BART
deep mixing
3-Layer MLP
GELU, nonlinear mixing
Pixel-Shuffle 4×
4×4 → 1 token
PaliGemma 2B → Action
Florence-2 Enc → Action
Llama-2 7B → Action
SmolLM2 360M → Action
CV = 0.055
Uniformly Critical
CV = 0.151
Threshold-Graded
CV = 0.215
Smooth Graded
CV = 0.314
Catastrophic Cliff
Increasing Fragility (CV: 0.055 → 0.314)